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ABSTRACT

Uniform metal contacts are critical for advanced thermoelectric devices. The uniformity of the contact resistance for gold, tungsten, and $SrRuO_3$ electrodes on polycrystalline ternary Bi_2Te_3-based alloys for different types of surface cleaning procedures was characterized. The presence of a nanometer-thick native oxide layer on the Bi_2Te_3 surface leads to large and non-uniform contact resistance. Surface treatments included solvent cleans and chemical and dry etching prior to metallization of the Bi_2Te_3. Only etching the surface led to a significant improvement in contact resistance uniformity. None of the tested contacts reacted with the underlying Bi_2Te_3 substrate. Etching resulted in the removal of the native oxide on the Bi_2Te_3 surface, which was characterized using X-ray photoelectron spectroscopy (XPS). The average thickness, chemistry, and dry etch rate of the native oxide was further characterized using XPS. The non-uniformity in contact resistance suggests that the native oxide grows non-uniformly on polycrystalline bismuth telluride surfaces.

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INTRODUCTION

Contact resistance can make a large contribution to the total resistance of a thermoelectric device, leading to Joule heating and a decrease in thermoelectric efficiency. An approximate expression for the effective figure of merit of a thin conductive metal on a thermoelectric material is $Z'/Z \sim (1 + 2\rho_C/L\rho)$, where Z' is the effective figure of merit with electrical contact resistance, ρ_C is the specific contact resistance, ρ is volume resistivity of the thermoelectric material, and L is the length of the thermoelectric device. Using $\rho \sim 1$ mΩ cm for Bi_2Te_3 and L = 1 cm, Z decreases by about 10% for ρ_C = 50 μΩ cm^2 and less than 1% for ρ_C = 1 μΩ cm^2. Given the present materials limitations to the figure of merit, thermoelectric energy conversion systems operate at the margins of efficiency and even relatively modest increases in contact resistance can unacceptably degrade device performance.

For this reason, many previous works have studied the effect of various surface treatments on contact resistance, in particular for binary bismuth telluride in thin film and bulk form, and for ternary materials (Bi, $Sb)_2Te_3$ (P-Bi_2Te_3) and $Bi_2(Te,Se)_3$ (N-Bi_2Te_3), which have the best thermoelectric properties for most applications below 400 K. For example, interfacial reactions,[1] ion implantation,[2] and surface functionalization[3,4] were used to improve the magnitude of contact resistance. Removing surface oxides also improves the contact resistance.[5]

The uniformity of contact resistance over a large area has not been addressed in thermoelectric devices. The ability to make uniform contacts over a large area is critical for device fabrication. In the present work, we characterize both the uniformity and magnitude of contact resistance for different types of surface treatment and for three different contact metals. N and P type polycrystalline ternary bismuth telluride alloys are studied. The uniformity of the

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contact resistance was controlled by exposing surfaces to different cleaning processes over a ~cm^2 area. Contact electrodes that had no interfacial reactions with the underlying substrates were selected, allowing better control over the interface homogeneity. An improvement in contact resistance uniformity was associated with the removal of native oxide on the surface of both N- and P-Bi_2Te_3 ternary materials. X-ray photoemission spectroscopy was used to characterize the chemistry and persistence of the surface oxide. The fact that the contact resistance was highly nonuniform for surfaces with a native oxide implies that ambient oxidation also occurs nonuniformly over these ternary polycrystalline alloys. This nonuniform oxide layer severely degraded the uniformity and magnitude of the contact resistance in these materials.

METHODS

Contact resistance measurements

The Cox and Strack (CS) method was used to extract specific contact resistance[6,7] and characterize uniformity. This method is more suitable for thin metal contacts on bulk materials and was successfully applied to bismuth telluride-based thermoelectrics.[8] We discuss the methods for uniformity characterization in the Results section. A series of circular metal contacts with 11 different diameters ranging from 50 to 550 microns was randomly patterned on polished surfaces of both types of thermoelectric coupons using a shadow mask. Each contact diameter was repeated another 11 times for a total of 121 measured contacts over the 1 cm^2 coupon area. Contact thickness was measured for a representative sample using a Dektak 150 profilometer. There was less than 2% variation in contact thickness for the different contact diameters. A Lakeshore 372 Resistance Bridge was used to measure the four-point probe resistance between each pad and the back surface at a current of 1 mA. Alternating current reversal at a frequency of ~13 Hz was used to null any thermoelectric voltages in the wires and the thermoelectric coupon.

Transmission electron microscopy

Scanning transmission electron microscopy (STEM) was used to investigate the morphology and composition of all Bi_2Te_3/metal interfaces. Electron transparent TEM samples were prepared using a dual beam SEM-FIB (FEI, Helios Nanolab 660). Samples were then analyzed with either a probe-corrected FEI Titan G2 80-200, or a probe-corrected FEI Titan Themis Z, both with a SuperX EDS detector. EDS spectral imaging was acquired as a series of frames where the same region was scanned multiple times. An electron probe of size less than 0.13 nm, convergence angle of 18.1 mrad, and current of ~75 pA was typically used for data acquisition. STEM high angle annular dark field (HAADF) images were recorded under similar optical conditions. Elemental maps were calculated from the EDS spectral imaging dataset with background subtraction. The EDS analysis was used to determine if interdiffusion of species occurred across the interfaces in these multilayer electrical contacts.

X-ray photoelectron spectroscopy

X-ray photoelectron spectroscopy (XPS) was performed with either a Kratos AXIS Supra or Kratos AXIS Ultra DLD instrument. Excitation in both instruments were from a monochromatic Al Kα (1486.7 eV) X-ray source operating at 75-150 W. The analyzed spots were an elliptical area of 300 x 700 microns. Survey spectra were recorded with 160 eV pass energy, while high resolution spectra were taken at 20 eV pass energy. Base pressures were less than $5x0^{-9}$ Torr. Charge neutralization was not required, however, low energy electrons supplied from a filament above the sample compensated for any possible differential charging effects. In the Kratos AXIS Supra, sputtering was performed with a Minibeam 6 ion gun utilizing monoatomic argon or argon clusters.

Metallographic polishing

Flat surfaces are essential for evaluation of contact resistance to reduce roughness and thus surface area. N- and P-Bi_2Te_3 were cut from ingots into 1 cm^2 flat plates of about 2 mm thickness and were polished using a Struers Tegramin-30 system. Coupons were polished with progressively finer grits from a 9 micron to sub-micron finish. A diamond DP-suspension was used for each finish. Coupons were washed between polishing steps with water to remove debris and avoid large scratches. After the final polish, coupons were rinsed with deionized water and dried with compressed air. Surfaces were then wiped with a one direction scrub in ethanol and dried with compressed nitrogen. Fourier transform infrared spectroscopy was performed with an ATR attachment to maximize surface sensitivity indicated that no residual organic compounds were present on the surfaces from the stated ingredients in the polishing compounds. Further cleaning steps were used for contact resistance analysis as described below.

Metal deposition

An Innotec model DS24C sputter system (E-10) was used for the DC magnetron sputter deposition of tungsten, and gold thin films. This system is cryopumped to a base pressure of 3×10^{-7} torr and includes three sputter sources mounted in a sputter down configuration. Sources were sputter cleaned for 5 minutes prior to deposition. Approximately 200 nm of tungsten and gold were deposited. Tungsten films were capped with gold to achieve better electrical contact with the probe tips. Sputter targets were ~20 cm x 1.25 cm diameter discs of 99.9+% purity with a working distance from sputter target to specimen of ~8.0 cm. Ultra-high purity argon (99.995%) was used for all sputter processes. Argon pressures of 5 and 14 mTorr were used for gold and tungsten, respectively. Substrate temperatures did not exceed 65C during deposition. Deposition rate was determined in separate experiments using a calibrated DEKTAK 3 (Veeco; Woodbury, NY) surface profilometer. The volume resistivity for the tungsten

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films was measured to be 7-8 μΩ cm, consistent with previous reports.[9]

$SrRuO_3$ films were prepared by 30° off-axis radio frequency (rf) magnetron sputtering from a 75 mm diameter $SrRuO_3$ sintered ceramic target. The target was acquired from Kurt J. Lesker Company. A custom sputter deposition system was used, with a rotating stage and a 75 mm sputter gun (Torus, Kurt J. Lesker Company) with a sputter down geometry. The distance between the target and substrate was set at 8 cm. Base pressures less than 10^{-6} Torr are achieved prior to deposition. The plasma was maintained under continuous flow of 5:1 argon and oxygen gas at 15 mTorr and at fixed flow rates. The films were verified to be metallic with a volume resistivity of 180 mΩ cm.

Summary of surface cleaning procedures

Prior to metallization, polished coupons of N and P-type materials were subjected to one of the following surface cleaning procedures. The roughness was measured for each cleaning recipe using a Bruker FastScan AFM. Coupons were loaded into the metal deposition chamber within 10 minutes of cleaning, with one exception indicated below. These different cleaning methods are designed to either only degrease the surface (SC-I, SC-II) or etch, i.e. remove material from the surface (SC-III, SC-IV).

SC-I: Samples were degreased in Lenium vapor for 15 minutes, followed by an acetone and 2-propanol rinse. Compressed N_2 gas was used to dry the samples. Lenium vapor is commonly used to remove oily substances, such as from fingerprints.

SC-II: Clean I was repeated, and the sample was then subjected to an oxygen plasma clean. A Harrick Plasma (PCD-01) plasma cleaner was used. The chamber was preconditioned using 100 W O_2 plasma for 15 minutes at 225 mTorr. The sample was then inserted into the chamber and cleaned under the same pressure and power conditions for 5 minutes. Tests showed that plasma clean times from 5 to 20 minutes yielded little difference in the resulting contamination. However, 20 minutes plasma cleaning heated the chamber up to 70C. A time of 5 minutes was chosen to avoid sample heating.

SC-III: Clean I was repeated, and the sample was immersed for 5 seconds in a 3:1:16 volumetric solution of HNO_3:HCl:H_2O. Nitric acid/hydrochloric acid mixtures have been previously reported to etch P and N bismuth telluride-based films.[10] The composition and time of the etch bath was chosen to remove no more than 100 nm of material from the surface, avoiding an increase in roughness. The etched surfaces had to be exposed to air briefly before metal deposition.

SC-IV: Clean I was repeated, then loaded into the sputter chamber. The sample was then RF sputter cleaned in-situ, i.e. under high vacuum without exposure to atmosphere prior to metallization, using ultrahigh purity Ar gas. Sputter cleaning involved application of 60 W RF power (200 V) to the platen holding the samples for an interval of time. Sputter times up to 15 minutes were studied. Over this range of time, the sputter erosion rate was found to be linear with Ar ion current. Rates of removal for P- and N-Bi_2Te_3 were ~50 nm/minute. SC-IV was only performed prior to gold and tungsten metallization owing to equipment configuration.

RESULTS

Figure 1 summarizes the extraction of contact resistance magnitude and uniformity using the Cox and Strack method

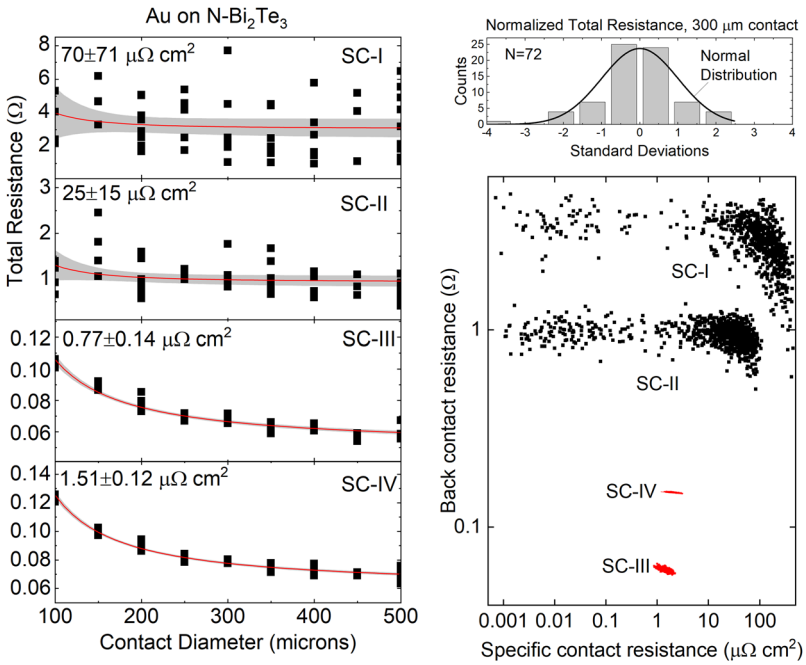


FIG. 1. Left panel: Total resistance versus contact diameter for different surface cleaning conditions. Square points refer to 4 probe measurements of the total resistance. The red curve is the best fit to the Cox and Strack equation. The shaded area refers to the 95% prediction bands for the fit for each set of data. Right panel: Bootstrap analysis for different Cox and Strack fitting procedures for the different surface cleaning conditions. The distribution in the total resistance for a given contact was approximately Gaussian (top right panel). The best fit parameters are plotted for 1000 different fits, assuming a Gaussian distribution of the measured resistance for each contact diameter (bottom right panel). The spread in the back contact resistance versus specific contact resistance indicates the uncertainty in the parameters and any correlation between said parameters. Contacts with lower contact resistance and more uniformity were achieved when the surface was etched (SC-III and SC-IV) rather than just removing surface contamination (SC-I, SC-II). The surface cleaning procedures are described in the text. Only N-Bi_2Te_3 material is shown for clarity.

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for gold contacts on N-Bi_2Te_3. The Cox and Strack (CS) method requires one to measure the resistance between a top contact of specified diameter and a large back contact made using silver paint, which ensures uniform current flow. The resistance between top contact and bottom contact is measured for several different diameters and fit to the following equation:

$$R = \frac{4\rho_C}{\pi d^2} + \frac{\rho}{\pi}\arctan\left(\frac{4t}{d}\right)\frac{1}{d} + R_B$$

Here, ρ is the bulk resistivity of the substrate material, d is the contact diameter, t is the coupon thickness, and R_B is the back contact resistance. The specific contact resistivity, ρ_C, and R_B are extracted as fit parameters. The coupon thickness and resistivity were both measured and held constant in the fits. The quasi-Newton method[11] was used to fit the data to this equation and extract best fit parameters and standard errors. In practice, to make this method work, a wide variety of diameters is needed for a high fidelity two parameter fit, where the small contact diameters are more sensitive to changes in contact resistance and larger diameters give a better estimate of the back contact resistance.

The left panel of Fig. 1 shows the CS results for the four cleaning procedures described in the Methods section for gold contacts on N-Bi_2Te_3. For SC-I and SC-II, the resistance for any given contact diameter was highly variable across the coupon and the total resistance reached values of several Ohms. Note that each data point at a given radius is a different metal contact located randomly across the ~1 cm^2 coupon area. Fitting the CS equation to this data (red line) results in a poor estimate of the contact resistance, indicated by the large standard error in specific contact resistance resulting from the fit and the large 95% confidence region (gray area), which accounts for the uncertainty in both specific contact resistance and back contact resistance fit parameters. For SC-III and SC-IV, the variability across the coupon is far smaller than for SC-I and SC-II. The difference between these cleaning procedures qualitatively indicates a difference in contact uniformity because the variability of the total resistance for any given diameter can be ascribed to the contact resistance as all contacts share the same back contact. The variability of multiple resistance measurements on a single contact was less than 1% of the total resistance. The variability within this interpretation persists up to the scale of the largest contact, which is 0.5 mm in diameter. We also measured the total resistance between four large 2 mm diameter contacts and the back contact, finding little variability for any given surface preparation, which bounds the length scale of the nonuniformity.

The errors in the fitted parameters, ρ_C and R_B, can be more precisely characterized by estimating their probability distribution using Monte Carlo-based bootstrapping methods,[11] as demonstrated in the right panel of Fig. 1. This estimate of the probability distribution can then be used to visualize the variability in fitted parameters ρ_C and R_B. We characterize the variability in total resistance for contacts

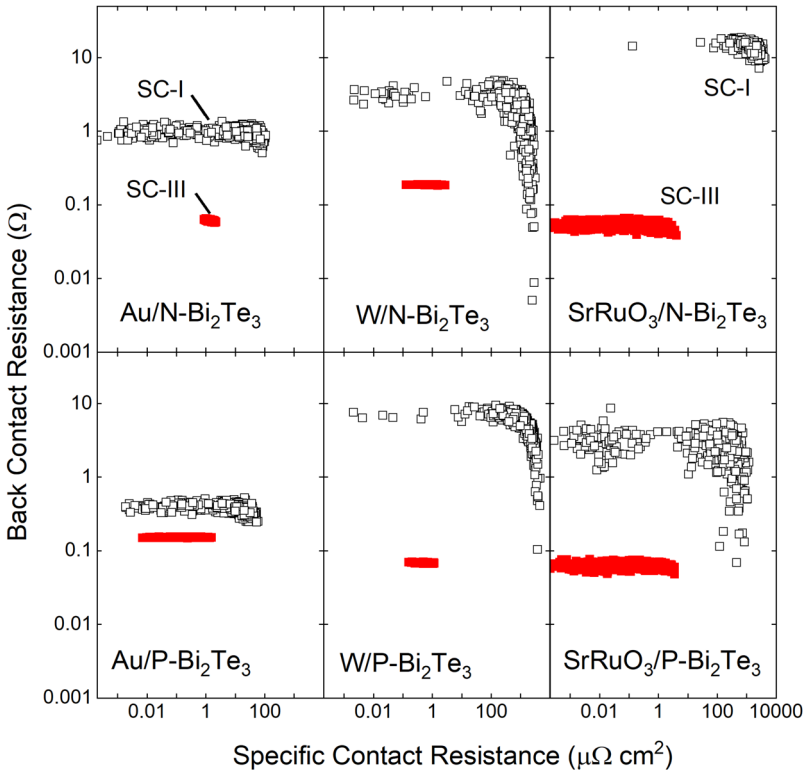


FIG. 2. Top panels: Bootstrap analysis, as described in Fig. 1 and the text, for two different surface cleans and 3 different contacts on N-Bi_2Te_3 coupons. Bottom panels: Bootstrap analysis for the same 3 contacts on P-Bi_2Te_3 coupons.

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of each diameter by extracting a standard deviation. The top panel shows these measurements for 72 separate data points at an example contact diameter of 300 microns. The probability distribution is approximately Gaussian, allowing a convenient definition of the standard deviation. Given this probability distribution defined experimentally, a new dataset is then simulated by drawing a random number from this probability distribution and a new set of parameters, ρ_C and R_B, is then extracted by fitting the simulated data to the CS equation. Repeating this procedure many times gives the confidence region in parameter space, quantifying the probability that a given set of parameters would occur given the variability in total resistance observed across a ~1 cm^2 ternary Bi_2Te_3 coupon.

Confidence regions for the different surface preparations used in this work are shown in the main right panel of Fig. 1. Each confidence region was constructed for N=1000 Monte Carlo simulations for each CS experiment. The standard error extracted from the fit procedure, which assumes a Gaussian distribution, was in rough agreement with the variation extracted from the confidence region for ~68% of data points. Plotting the confidence region this way allows a direct comparison of the uncertainty without a priori assuming any particular probability distribution in the fitted parameters.[11]

The main panel on the right of Fig. 1 indicates a large variability in contact resistance for SC-I and SC-II. SC-III and SC-IV have far lower variability and lower values of ρ_C and R_B. The confidence regions for SC-I and SC-II suggest that in a small percentage of cases, a CS experiment would yield a very small estimate of contact resistance that originates from the large variability in contact resistance across the test coupon. Characterizing the variability by estimating the probability distribution of the fitted parameters better quantifies the qualitative conclusions from the left panel of Fig. 1 and allows one to directly compare the fitted parameters ρ_C and R_B for different samples. This method of analysis was used for comparing contact variability for the remainder of the manuscript.

We analyze the variability in contact resistance established in Fig. 1 for three different metals: gold, tungsten, and amorphous $SrRuO_3$ in the left, middle, and right columns of Fig. 2. The top and bottom rows of Fig. 2 compare the analysis for N- and P-Bi2Te3, respectively. Two different surface cleaning procedures are shown, confirming the results of Fig. 1 for tungsten and $SrRuO_3$. SC-III improves contact resistance and uniformity for these other contacts versus SC-I. The other surface cleaning procedures reproduce the results in Fig. 1 and 2 and are omitted for clarity. Fig. 2 also shows how surface clean SC-I leads to a large variation in contact resistance for different depositions, in addition to a large variation over the area of a single coupon. This variation for different deposition runs is removed when using an appropriate surface clean, in this case SC-III. For all the contact metals studied, cleaning the surface from contaminants using a degreasing method (SC-I, SC-II) was not sufficient to get a low and uniform contact resistance. SC-III and SC-IV were confirmed to remove material from the surface of the coupon, and in all cases led to an improvement in contact uniformity.

To understand the observations in Figure 2, the contact-Bi_2Te_3 interface was characterized using cross-sectional TEM for different surface cleaning procedures. In Figure 3 we show representative STEM-HAADF cross sectional images for gold and $SrRuO_3$ contacts on P-Bi_2Te_3 and tungsten on N-type Bi_2Te_3, which were subjected to SC-I. In each image, a sharp interface between the Bi_2Te_3 substrate and the metal layer is observed. We overlay EDS maps across the interface, where the color intensity corresponds to the x-ray signal intensity from relevant constituent elements in metal and Bi_2Te_3 layers. The TEM observations show no morphological changes and no discernable interdiffusion or second phase formation at the metal-Bi_2Te_3 interface. Figure 3 rules out metal-Bi_2Te_3 interdiffusion and surface roughness as mechanisms for explaining the improvement in electrical inhomogeneity demonstrated in Figure 1 and 2 when the surface is etched. AFM measurements of roughness were in the 10 nm range over a scan area of about 5 mm^2 with no systematic variation in roughness across cleaning procedures, consistent with the TEM results.

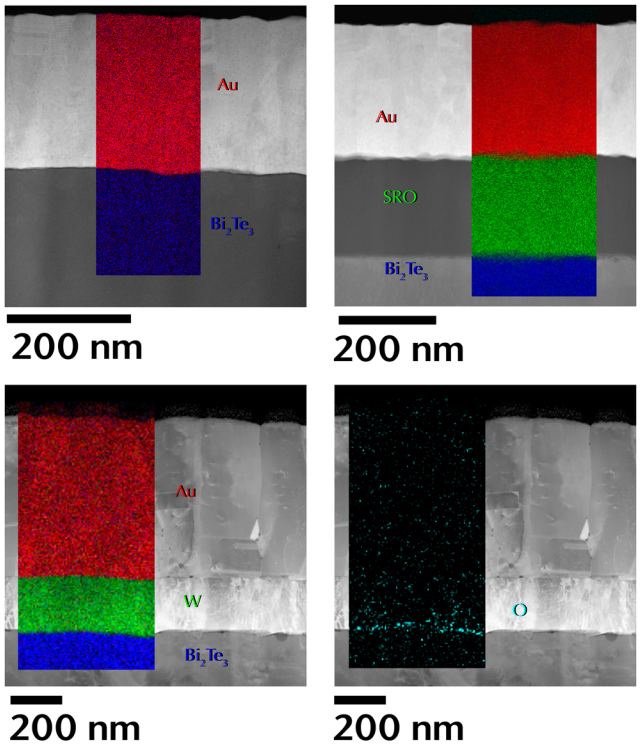


FIG. 3. STEM-HAADF cross-sectional images for gold on P-Bi_2Te_3, tungsten on N-Bi_2Te_3, and $SrRuO_3$ on P-Bi_2Te_3. The Bi_2Te_3 surface was only degreased (SC-I) prior to metal deposition. The EDS maps are overlaid on the HAADF images in composite color maps, where Red is gold, Blue is tellurium, Green is either tungsten or strontium and Cyan is oxygen. The bismuth map is omitted for clarity, but coincides with that for tellurium. The titanium map that coincides with the strontium map is also omitted. There is no obvious chemical interdiffusion or interfacial intermetallic formation between the different layers. There is little roughness at the interface between the metal and substrate. Oxygen is detected at the metal-Bi_2Te_3 interface.

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Oxygen was detected at the metal-Bi_2Te_3 surface, as shown in the Cyan EDS map for the W-Bi_2Te_3 interface. The presence of oxygen is suggestive of a native oxidation process and forms a primary hypothesis for the variability in contact resistance measurements presented in Fig. 1 and 2.

TEM views the samples in cross section, so there is very little amount of material sampled right at the Bi_2Te_3 surface and so it is difficult to study oxidation in a quantitative and systematic manner. XPS measurements were used to sample a larger area and to analyze the surface chemistry for the different surface preparations with a higher surface sensitivity to the first few atomic layers in the Bi_2Te_3. Figure 4 shows the XPS spectra for SC-II, SC-III, and SC-IV, corresponding to a Lenium/plasma treatment, a wet nitric acid etch, and the dry argon ion etch, respectively. All samples were exposed to air for 10 minutes prior to XPS analysis. No additional surface treatment was conducted in the XPS chamber prior to collecting spectra. Each surface will therefore have some amount of oxidation in air, but the cleaning methods can be better compared to one another on a relative basis. Figure 4 shows the Bi 4f peaks from exposed Bi_2Te_3 regions on the coupons with different surface cleans. Surface oxidation is indicated by the high binding energy component on both the Bi 4f 7/2 and Bi 4f 5/2 peaks. Te 3d peaks are not shown, but tellurium oxidation follows similar trends to that observed for bismuth. The results are summarized in the bar graph of Figure 4 where the percentage of the peak areas are shown for oxidized species. For the N-type material, there was some discrepancy between the oxidation observed in the Bi 4f versus Te 3d for the SC-III and SC-IV cleans: however, the overall results are still qualitatively similar.

XPS experiments require the collection of photoelectrons over a large region of several hundred microns to improve the signal to noise ratio. Therefore, the peak area metric for the oxide relative to the main peak is an indicator of either the thickness of a homogeneous overlying oxidized layer, or the volume fraction of oxidized species dispersed across the oxidized area. Figure 4 shows that SC-II had the most oxidized species, followed by SC-IV, and finally SC-III, which had the least oxidation. Comparing these results to Fig. 1 and 2, we conclude that the surfaces with less oxide have a lower and more uniform contact resistance.

To more quantitatively characterize the oxidation process, in situ Ar ion sputtering on as-polished surfaces was performed under UHV conditions. Figure 5 shows XPS spectra for Te 3d and Bi 4f peaks for N- and P-Bi_2Te_3 after different etching times. Sb 3d peaks were also analyzed for P-Bi_2Te_3 samples. Traditional monoatomic Ar sputtering at 5 keV and Ar cluster sputtering were used. Cluster sputtering energies of 20 eV/Ar was performed to reduce the possibility of differential sputtering effects. The cluster interacts with a greater number of surface atoms relative to monoatomic sputtering due to the difference in size and also distributes energy more evenly to the surface resulting in less energy per ion. After etching to remove all oxide signal from Bi 4f, Te 3d, and Sb 3d peaks, the coupons were exposed to air for 60 seconds and reinserted into the vacuum chamber. Oxide peaks are seen after this exposure to air, confirming that the oxidation

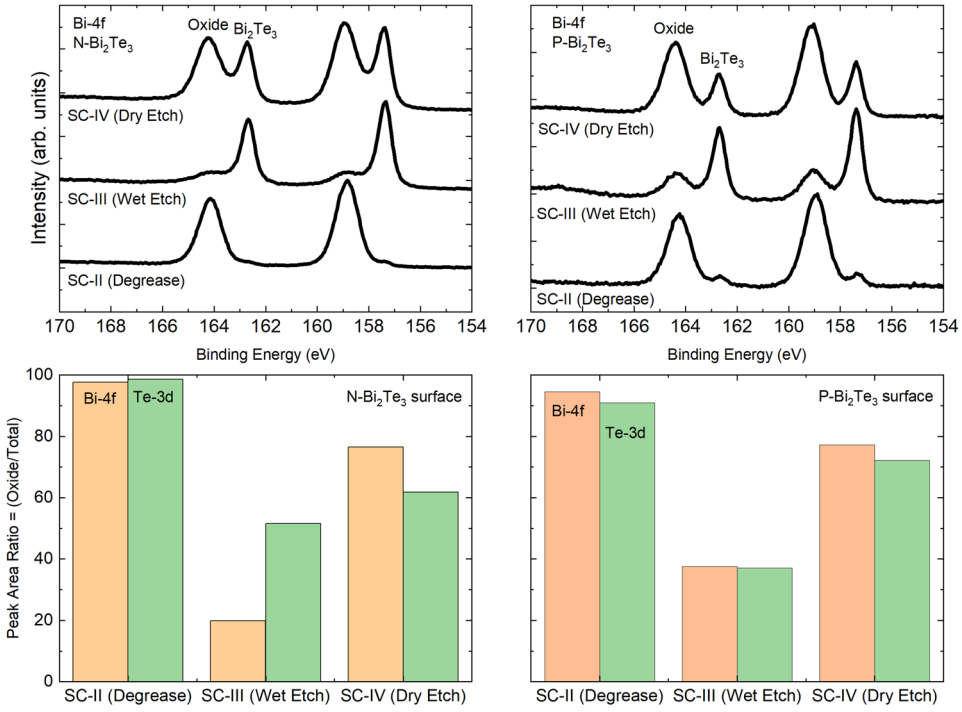


FIG. 4. Top panels: XPS spectra for N- (left) and P- (right) Bi_2Te_3 surfaces after the surface treatments indicated. SC-II was a degreasing process that only removes contaminants from the surface. SC-III and SC-IV were wet and dry etch processes as described in the text. Lower panels: Peak area ratio for the oxide versus elemental peaks for Bi-4f and Te-3d, as indicated, for the different surface cleaning processes on N- (left) and P- (right) Bi_2Te_3. Each cleaning procedure was followed by ~10 minutes exposure to air. SC-II has the greatest amount of oxide on the surface. The wet etch (SC-III) shows the least amount of oxide.

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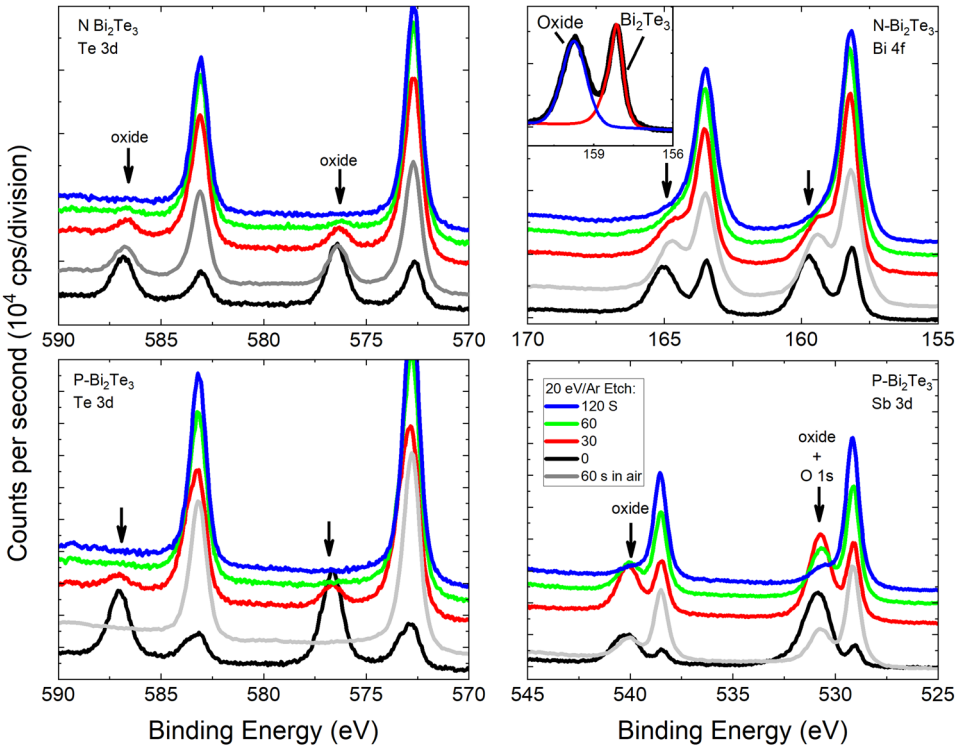


FIG. 5. X-ray photoelectron spectroscopy (XPS) of N- and P-Bi_2Te_3 coupon surfaces. The as received surface after the standard clean was analyzed (0 second etch time – black lines). The surface was slowly sputtered away with Ar ions at 20 eV/ion and the XPS spectra were remeasured after 30 seconds (red lines), 60 seconds (green lines), and 120 seconds (blue lines). The change in the oxidation peaks for Te 3d (left panel), Bi 4f (upper right panel), and Sb 3d (lower right panel) peaks with sputter time is shown. After the sputtering experiment, the sample was exposed to air for 60 seconds and XPS spectra were retaken. The oxide peaks (indicated by arrows) decrease with sputtering time for each peak. After exposure to air for 60 seconds, each oxide peak, with the exception of the P-Bi_2Te_3 tellurium peaks, return to about the same intensity as the initial condition, revealing that the oxide is native to N- and P-Bi_2Te_3. The inset of the upper right panel shows the deconvolution of oxide and intrinsic bismuth telluride peaks using Gaussian peak shapes that is used to quantitatively analyze the apparent oxide thickness in Fig. 6.

process is native to the P- and N-Bi_2Te_3 surfaces under ambient conditions.

The peak areas for each of the curves in Figure 5 were used to calculate an apparent oxide thickness following the procedure outlined by Strohmeier for aluminum oxide on aluminum.[12] Inelastic mean free path values were obtained from the NIST Standard Reference Database 71 for Bi_2O_3, TeO_2, Sb_2O_3, and Bi_2Te_3.[13] The density of Bi_2O_3, TeO_2, and Sb_2O_3 were used as the representative oxide species. It is unclear whether the native oxide forms in these discrete forms. However, comparing the spatially-averaged thickness this way gives insight into the chemistry of the native oxide layer during the etching process. The apparent oxide thickness defined this way for bismuth and tellurium signals is similar in the as prepared state. For N-Bi_2Te_3, the etch rate of both bismuth and tellurium oxidation signals is similar for two different beam energies (5 keV/Ar and 20 eV/Ar) for the first few seconds of etching. The corresponding signals in the P-Bi_2Te_3 etch more quickly at 5 keV/Ar than 20 eV/Ar. The presence of antimony in the P type material seems to cause preferential sputtering of bismuth and tellurium using monoatomic sputtering. For cluster sputtering, the profiles obtained for bismuth oxide and tellurium oxide are for N- and P-Bi_2Te_3. The Sb 3d peak overlaps the O 1s, but the antimony oxide component can clearly be extracted by using the Sb 3d 3/2 peak. The oxide component for antimony shows a larger initial thickness than that for bismuth or tellurium. The antimony oxide component also etches slower than the bismuth and tellurium oxides. The etch rate of the antimony oxide is independent of the ion beam energy 20 eV/Ar and 5 keV/Ar. Upon reoxidation of the N-Bi_2Te_3 both the bismuth and tellurium components show similar oxide thickness. There is a small difference in the thickness depending on the sputtering condition, which could be due to small differences in surface morphology induced by the bombarding ion beam. In contrast, for the P-Bi_2Te_3, bismuth and tellurium show almost no reoxidation in air, while antimony shows oxidation. The antimony oxide peak area for P-Bi_2Te_3 is similar to that of bismuth and tellurium in the N-Bi_2Te_3. This latter observation could suggest that there is a differential sputtering between antimony and bismuth/tellurium that leaves behind an excess of antimony, which subsequently oxidizes in air and passivates the surface.

DISCUSSION

For polycrystalline semiconductor-metal interfaces, inhomogeneity at the interface is usually assumed to lead to poor electrical properties.[14] In the present study, variability in contact resistance is the observation that is used to quantify inhomogeneity for different surface cleaning procedures. XPS experiments were then used to tie this observed inhomogeneity to native oxidation of the surface under ambient conditions. There was no difference in roughness, as characterized using AFM, for the etched versus solvent cleaned surfaces studied here. The cross sectional TEM results of Figure 3 showed no difference in morphology and the absence of interdiffusion for all surface cleaning procedures.

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Thus, the main result presented here is that contact resistance not only improves, but becomes more uniform when the native surface oxide is removed, which could only be accomplished by etching away surface material using the surface cleaning procedures SC-III or SC-IV, corresponding to wet or dry etch processes, respectively. Associating the nonuniformity of contact resistance with surface oxidation using the XPS evidence, we further conclude that the surface oxidation process is non-uniform. There could be areas on the coupon that have no oxidation, different native oxide thicknesses, and/or different oxidative species, all of which would be difficult to discriminate in an XPS experiment. Compositional measurements in the TEM with either EDS (shown here) or electron energy loss spectroscopy could be used to measure the thickness of the oxide layer in these samples. However, because the inhomogeneity is over a length scale of several hundred microns judging from the contact resistance results, it will be difficult to quantitatively measure the range of thickness with TEM alone given the limited size of an electron transparent region in a TEM sample and the need for numerous samples to get a good statistical sampling of the interfacial inhomogeneity.

Oxidation of the pure bismuth telluride and bismuth selenide surfaces has been studied previously. Bi_2Te_3 has been observed to oxidize slowly under ambient conditions, where "ambient conditions" refers to an air environment with an unspecified humidity. A few nm oxide layer forms within a day under ambient conditions and further oxidation seems to occur on longer time scales (~months), which suggests passivation from a native oxide.[15] Bi_2Se_3 also forms a native oxide within a few days under ambient conditions.[16] Surface oxidation under ambient conditions was also observed in $Bi_{0.5}Sb_{1.5}Te_3$ particles.[17] In contrast to ambient conditions, Bi_2Te_3 and Bi_2Se_3 were observed to be resistant to oxidation when exposed to individual oxidizing species. Water and oxygen separately have a low sticking coefficient on bismuth telluride cleaved surfaces with well-defined crystallographic planes.[18] Bi_2Se_3 was reported to be resistant to oxidation in pure O_2 or NO_2,[19] consistent with Ref. 18. Bi_2Te_3 was reported to be resistant to oxidation in pure water vapor and O_2.[20] This difference in oxidation for bismuth telluride in mixed ambient atmospheres versus pure O_2 or water vapor remains obscure, but is reminiscent of the difference between wet and dry oxidation in silicon.[21] In that context the oxidant solubility in SiO_2 explains the much higher oxidation rate for wet oxidation. The question of how an inhomogeneity in oxidation can arise is now discussed.

In general, native oxide layers tend to be inhomogeneous when there are multiple stable oxide phases.[22,23] Variation in electrical properties could result if these multiple oxides had different electrical properties, oxidation rates, or both. The Bi_2Te_3-TeO_2 pseudobinary phase diagram reveals the presence of multiple oxide phases containing both bismuth and tellurium as well as Bi_2O_3 and TeO_2[24] The Sb_2Te_3-TeO_2 psuedobinary phase diagram, relevant for P-Bi_2Te_3, shows a similar phase diagram.[25] From a phase stability point of view, these different phases would notionally coexist depending on the prevailing oxygen partial pressure. Density functional

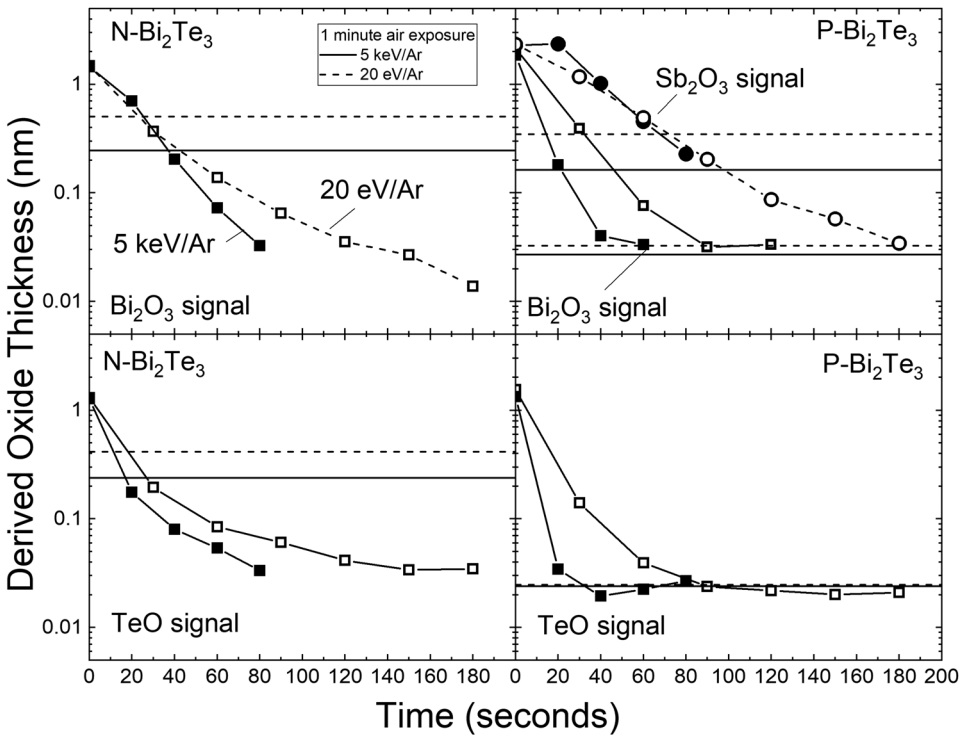


FIG. 6. Derived oxide thickness from the XPS spectra shown in Fig. 5 as a function of sputtering time, at Ar ion energies of 5 keV/ion (closed symbols) and 20 eV/ion (open symbols). The dashed lines indicate the derived oxide thickness after exposure to air for 60 seconds for 5 keV/ion (solid lines) and 20 eV/ion (dashed line). The P-Bi_2Te_3 data in the upper right panel shows the bismuth oxide signal (squares) and the antimony oxide signal (circles). The P-Bi_2Te_3 sputter rates are more energy and species dependent than the N-Bi_2Te_3.

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calculations of oxygen in equilibrium with Bi_2Te_3 show that bismuth has a greater oxygen reactivity than tellurium, which would favor the formation of Bi_2O_3 on the surface of bismuth telluride.[26] These authors also show that if the bismuth telluride surface was terminated with tellurium rather than bismuth, oxidation would be much slower. The latter conclusion is consistent with an observation[27] that tellurium terminated surfaces of pure Bi_2Te_3 show a passivation to oxidation. In the present study, Figures 5 and 6 seem only to show a preponderance for antimony oxidation in P-Bi_2Te_3, while bismuth and tellurium show a similar apparent oxide thickness after air exposure for N- and P-Bi_2Te_3, which neither supports nor refutes the surface termination hypothesis presented in Ref. 26. Nonetheless, surface termination is difficult to control in the polycrystalline specimens studied here and could therefore be another source of inhomogeneous oxidation. Given an estimated grain size of ~1-10 microns and contact diameters of 50 – 500 microns, we would not expect oxidation at grain boundaries alone to produce the observed effect in Figure 1.

CONCLUSIONS

A large inhomogeneity in electrical contact resistance was associated with the presence of a thin native oxide on P- and N-Bi_2Te_3 surfaces. Oxidation occurred under ambient atmospheric conditions and could be readily removed using wet or dry etching, which improved the electrical homogeneity. Three different metals were used to assess electrical properties. XPS measurements were necessary to detect and characterize the presence of the oxide, which must be assumed to occur inhomogeneously over the surface given the electrical contact resistance measurements. The nature of this inhomogeneity remains obscure but points to an important difficulty in utilizing the surface-mediated electronic properties of ternary chalcogenide materials in various applications.

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REFERENCES

[1] R. P. Gupta, O. D. Iyore, K. Xiong, J. B. White, K. Cho, H. N. Alshareef, and B. E. Gnade, "Interface characterization of cobalt contacts on bismuth selenium telluride for thermoelectric devices," Electrochemical and Solid-State Letters **12**(10), H395–H397 (2009).

[2] P. J. Taylor, J. R. Maddux, G. Meissner, R. Venkatasubramanian, G. Bulman, J. Pierce, R. Gupta, J. Bierschenk, C. Caylor, and J. D'Angelo, "Controlled improvement in specific contact resistivity for thermoelectric materials by ion implantation," Applied Physics Letters **103**(4), 043902 (2013).

[3] T. Cardinal, M. Kwan, T. Borca-Tasciuc, and G. Ramanath, "Multifold electrical conductance enhancements at metal–bismuth telluride interfaces modified using an organosilane monolayer," ACS Applied Materials & Interfaces **9**(3), 2001–2005 (2017).

[4] T. Cardinal, Devender, T. Borca-Tasciuc, and G. Ramanath, "Tailoring electrical transport across metal–thermoelectric interfaces using a nanomolecular monolayer," ACS Applied Materials & Interfaces **8**(7), 4275–4279 (2016).

[5] R. P. Gupta, K. Xiong, J. B. White, K. Cho, H. N. Alshareef, and B. E. Gnade, "Low resistance Ohmic contacts to Bi_2Te_3 using Ni and Co metallization," Journal of The Electrochemical Society **157**(6), H666–H670 (2010).

[6] R. H. Cox and H. Strack, "Ohmic contacts for GaAs devices," Solid-State Electronics **10**(12), 1213–1218 (1967).

[7] D. K. Schroder, *Semiconductor material and device characterization*, John Wiley & Sons, 2006.

[8] R. P. Gupta, J. B. White, O. D. Iyore, U. Chakrabarti, H. N. Alshareef, and B. E. Gnade, "Determination of contact resistivity by the Cox and Strack Method for metal contacts to bulk bismuth antimony telluride," Electrochemical and Solid-State Letters **12**(8), H302–H304 (2009).

[9] P. Petroff, T. T. Sheng, A. T. Sinha, G. A. Rozgonyi, and F. B. Alexander, "Microstructure, growth, resistivity, and stresses in thin tungsten films deposited by RF sputtering," Journal of Applied Physics **44**(6), 2545–2554 (1973).

[10] L. M. Goncalves, J. G. Rocha, C. Couto, P. Alpuim, M. Gao, D. M. Rowe, and J. H. Correia, "Fabrication of flexible thermoelectric microcoolers using planar thin-film technologies," Journal of Micromechanics and Microengineering **17**(7), S168 (2007).

[11] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical recipes 3rd edition: The art of scientific computing* (Cambridge University Press, 2007).

[12] B. R. Strohmeier, "An ESCA method for determining the oxide thickness on aluminum alloys," Surface and Interface Analysis **15**(1), 51–56 (1990).

[13] A. Jablonski, NIST Electron Inelastic-Mean-Free-Path Database, National Institute of Standards and Technology, Gaithersburg Google Scholar, 2010.

[14] R. T. Tung, "Electron transport at metal-semiconductor interfaces: General theory," Physical Review B **45**(23), 13509–13523 (1992).

[15] H. Bando, K. Koizumi, Y. Oikawa, K. Daikohara, V. Kulbachinskii, and H. Ozaki, "The time-dependent process of oxidation of the surface of Bi_2Te_3 studied by x-ray photoelectron spectroscopy," Journal of Physics: Condensed Matter **12**(26), 5607 (2000).

[16] D. Kong, J. J. Cha, K. Lai, H. Peng, J. G. Analytis, S. Meister, Y. Chen, H.-J. Zhang, I. R. Fisher, Z.-X. Shen, and Y. Cui, "Rapid surface oxidation as a source of surface degradation factor for Bi_2Se_3," ACS Nano **5**(6), 4698–4703 (2011).

[17] Y. Zhao and C. Burda, "Chemical synthesis of $Bi_{0.5}Sb_{1.5}Te_3$ nanocrystals and their surface oxidation properties," ACS Applied Materials & Interfaces **1**(6), 1259–1263 (2009).

[18] D. Haneman, "Adsorption and bonding properties of cleavage surfaces of bismuth telluride," Physical Review **119**(2), 567–569 (1960).

[19] V. A. Golyashov, K. A. Kokh, S. V. Makarenko, K. N. Romanyuk, I. P. Prosvirin, A. V. Kalinkin, O. E. Tereshchenko, A. S. Kozhukhov, D. V. Sheglov, S. V. Eremeev, S. D. Borisova, and E. V. Chulkov, "Inertness and degradation of (0001) surface of Bi_2Se_3 topological insulator," Journal of Applied Physics **112**(11), 113702 (2012).

[20] L. V. Yashina, J. Sánchez-Barriga, M. R. Scholz, A. A. Volykhov, A. P. Sirotina, V. S. Neudachina, M. E. Tamm, A. Varykhalov, D. Marchenko, G. Springholz, G. Bauer, A. Knop-Gericke, and O. Rader, "Negligible surface reactivity of topological insulators Bi_2Se_3 and Bi_2Te_3 towards oxygen and water," ACS Nano **7**(6), 5181–5191 (2013).

[21] B. E. Deal and A. S. Grove, "General relationship for the thermal oxidation of silicon," Journal of Applied Physics **36**(12), 3770–3778 (1965).

09 May 2026 08:04:42

© Author(s) 2019

[22]G. Hollinger, E. Bergignat, J. Joseph, and Y. Robach, "On the nature of oxides on InP surfaces," Journal of Vacuum Science & Technology A **3**(6), 2082–2088 (1985).
[23]G. Hollinger, J. Joseph, Y. Robach, E. Bergignat, B. Commère, P. Viktorovitch, and M. Froment, "On the chemistry of passivated oxide–InP interfaces," Journal of Vacuum Science & Technology B: Microelectronics Processing and Phenomena **5**(4), 1108–1112 (1987).
[24]P. Schmidt, O. Bosholm, and H. Oppermann, "Investigations of the ternary system Bi/Te/O. II. The thermal behaviour and the relations of phase coexistences of Bi~ 2O~ 2Te," Zeitschrift Fur Naturforschung B **52**(12), 1461–1466 (1997).
[25]L. A. Demina, D. V. A., S. L. Buzhak, B. A. Popovkin, and A. V. Novoselova, "Certain sections of the oxygen-antimony-tellurium system," Russ. J. Inorg. Chem. **26**, 771–773 (1981).
[26]M. Denis, C. Keke, S. Paul, N. B. Felix, H. Martin, H. Steffen, J. P. Peter, S. Till, and W. Cedric, "On atomic mechanisms governing the oxidation of Bi_2Te_3," Journal of Physics: Condensed Matter **29**(48), 485705 (2017).
[27]F. Fang, R. L. Opila, R. Venkatasubramanian, and T. Colpitts, "Preparation of clean Bi_2Te_3 and Sb_2Te_3 thin films to determine alignment at valence band maxima," Journal of Vacuum Science & Technology A: Vacuum, Surfaces, and Films **29**(3), 031403 (2011).

09 May 2026 08:04:42

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